European Developer Conference



Presented By:
Name, Title



Treo 600

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Agenda

- **Treo 600 Overview**
- **Technical Details:**
  - SDK
  - 5-way Navigation
  - Power Management

  - Telephony
  - Network Connectivity
  - SMS & NBS Messages

Treo 600 Overview

- **Treo 600 Design Goals**
  - Have a phone-like look and feel
  - Maintain the award-winning Treo functionality
  - Add breakthrough features based on past learning
- **Revolutionary size for Smartphones**
- **Innovative new keyboard design**
- **Unique hardware & software integration**



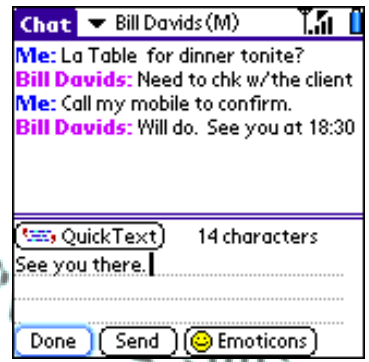
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Treo 600: Hardware & Software Integration

- **Better navigation**
  - 5-Way with Palm OS 5
- **Better messaging**
  - Keyboard w/ threaded SMS
- **Better multimedia experience**
  - Integrated camera for taking and sending pictures
  - Picture Caller ID







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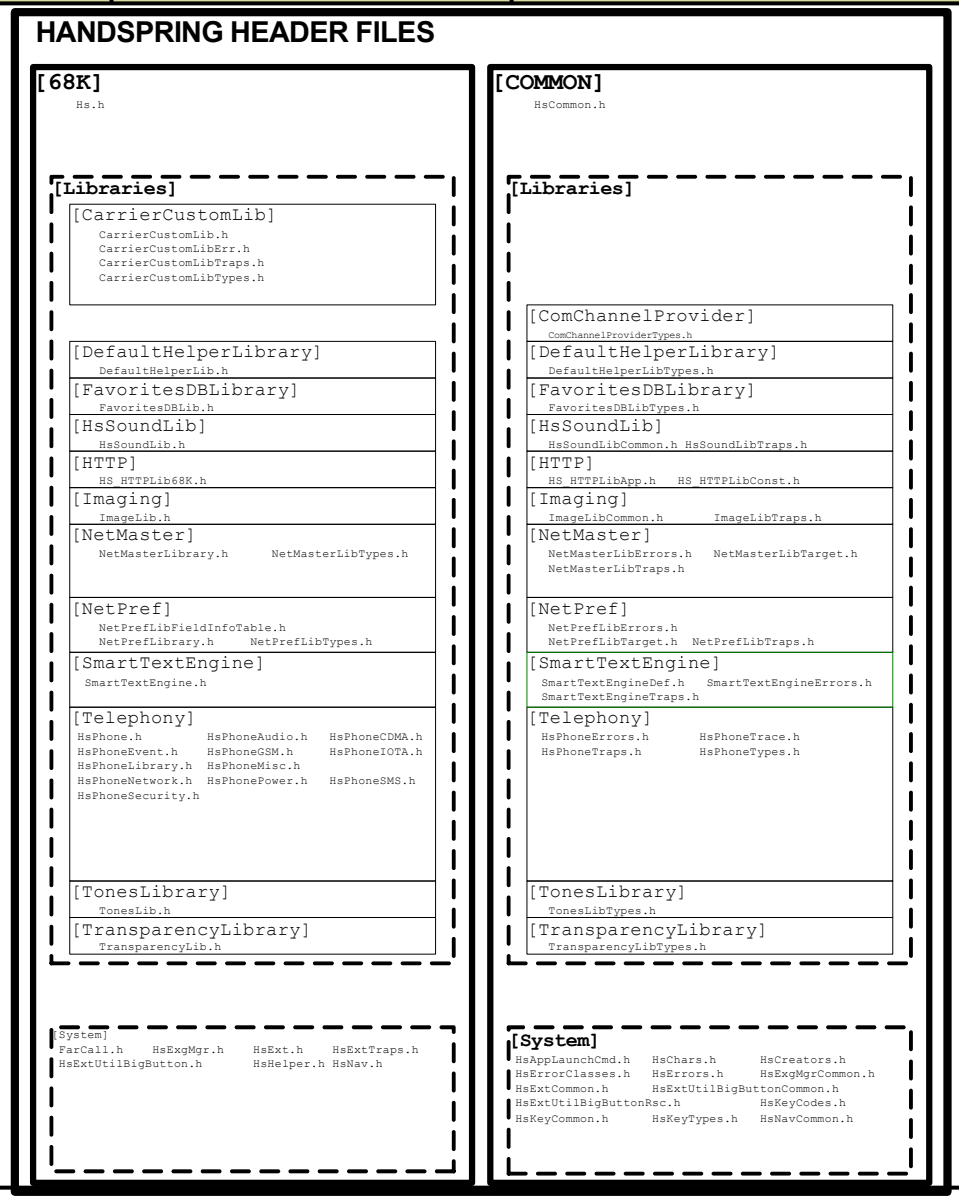
SDK

- Header files
- Simulator
- Sample Code
- Software Tools
- Developer Guide

palmsource

SDK: Header Files

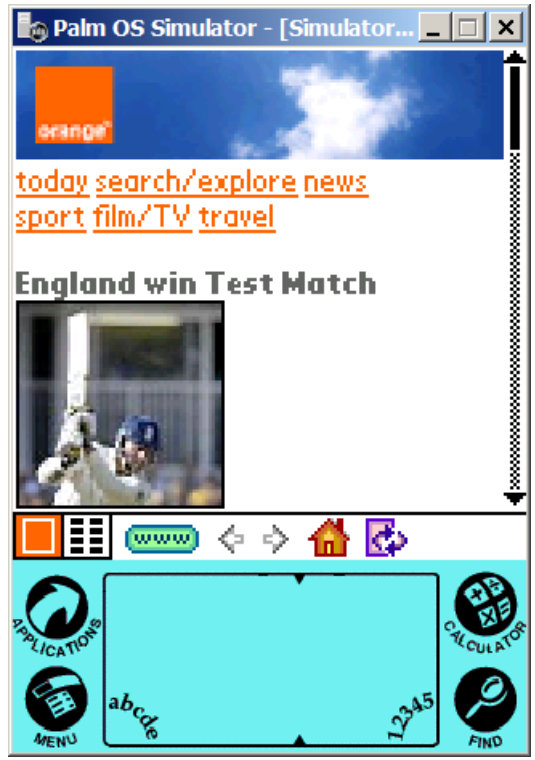
- **Simply include 68K/Hs.h**
  - It includes everything else
  - Include path is Handspring5
- **Module for each Library**
- **It's an exact copy of our internal headers w/o the private ones:**
  - Less errors
  - More up-to-date Headers
- **Easy to publish ARM**



palm POWERED

SDK: Palm Simulator

- **Easy to test network access:**
  - Settings -> Communication -> Redirect NetLib calls to Host TCP/IP
- **Use the Debug Simulator! It will help you find and fix:**
  - memory leak
  - Un-freed chunks
  - and other issues!
  - It's will available on our website next week
- **Don't forget to set the Radio Type**
  - Settings -> GSM Radio
  - Settings -> CDMA Radio



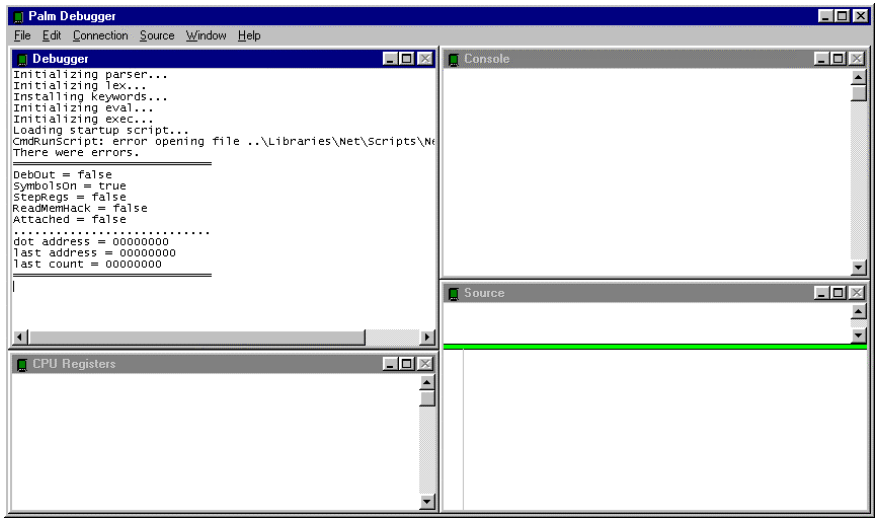
SDK: Sample Code

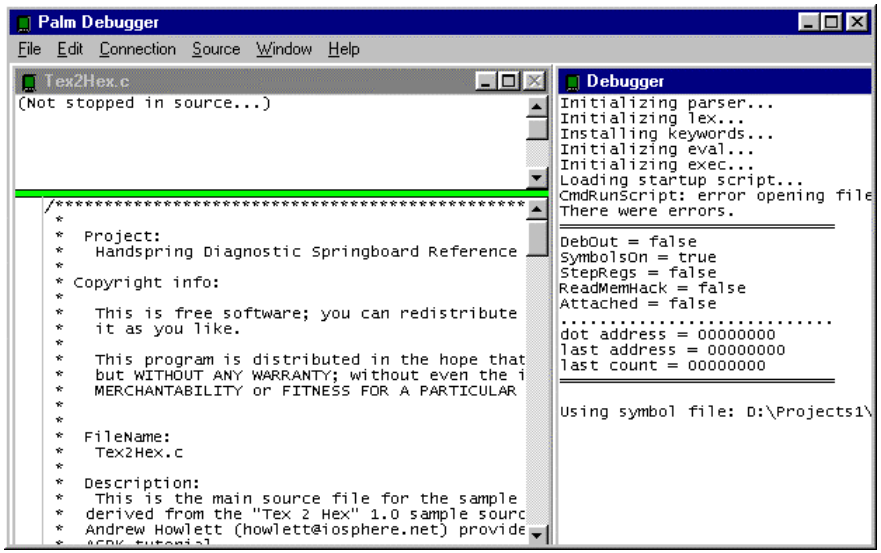
- **5-way Navigation: HsNavSample.zip**
- **HTTP Library: HTTPLibTest.zip**
- **Phone Library: PhoneInfo.zip**
- **SMS Library: SimpleSMS.zip**
- **Smart Text Engine: STETest.zip**

Don’t forget: This is a Palm OS device so most Palm OS sample code on Palm Source's website are good examples

SDK: Software Tools

- **NONE except Palm-Debugger**
  - We work with:
    - Standard GCC or
    - Metrowerks Development Tools
- **With Palm-Debugger**
  - You can:
    - Download application on the device
  - For GCC code:
    - look at source code
    - Set break point in your source code
  - For all other code:
    - assembly
    - see register values
  - And more
- **Serial cable only!**





SDK: Developer Guide

- Introduction & Overview:
  - General + Application
  - SDK Overview
- Audio & Tones
- Data Communication
  - NetPref
  - NetMaster
  - HTTP
- Telephony
  - Events
  - Types & Enumerations
    Network, Audio, CDMA, GSM, IOTA, Security, Others
  - SMS Library: SMS & NBS
- Imaging
  - Camera Library
  - Image Library
- System Extension
  - 5-Way Navigation
  - Smart Text Engine
  - Handspring Extensions
  - Transparency Library
  - Default Apps Preference API
  - Exchange Manager Support
  - REM sleep mode
- Debugging on Treo
  - Configuration
  - Source Level Debugging
  - Low-level debugging
- Electrical & Mechanicals
  - Cradle Connectors
  - Mechanical files on the web

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5-Way Navigation…



- **Critical to a good user experience**
  - Internal applications have been optimized for one-handed operation
  - New usage model requires 5-way navigation
  - It's a major portion of one handed navigation strategy
  - If possible application should not depend on the stylus
    - Example: Card game
- **It's two-dimensional: Left-Right, Up-Down, No cycling**

- **It's different than Palm Solutions Group ☹**
- **It's supported by Palm Source ☺**
  - Their latest Header supports it (R3) ☺

… 5-Way Navigation

- **Form receives frmObjectFocusTake and frmObjectFocusLost**
- **When frmObjectFocusTake is *handled*: Object focus mode**
  - 5-way keys are used for navigation
    - vchrRockerUp, vchrRockerDown, vchrRockerLeft, vchrRockerRight, vchrRockerCenter
- **When frmObjectFocusTake is *not handled*: Application focus mode**
  - 5-way up and down keys are used as page up and down keys
    - vcharPageUp and vcharPageDown

- **The system tries to do 5-way if it does not find a resource for it**
- **But you really need an "fnav" resource in your rcp file to make it work well**
  - Form assignment: Identifies the form it applies to
  - Header: Version, Number of elements in the object list
  - Object List: Objects in tab order with the above and below object
- **It has to be in the same code resource as your form!**

//Form Assignment **USING Palm-RC (Handspring's PilRC)**

HEX "fnav" ID mainFormID

// **Header**

```
1.w                          // Resource version
2.w                          // Objects in focus order
20.w                         // Size of header
8.w                          // Size of list elements
0x00000001.l                 // navFlags = kFrmNavHeaderFlagsObjectFocusStartState
mainFormTableID.w            // ID of initial focus object
mainFormDetailsButtonID.w    // ID of jump to object
mainFormDetailsButtonID.w    // ID of bottom-left object
0.w                          // Padding
```

// **Objects in tab order**

```
mainFormTableID.w            // ID of object
0.w                          // objectFlags = none
0.w                          // ID of above object = none
mainFormDetailsButtonID.w    // ID of below object

mainFormDetailsButtonID.w    // ID of object
0.w                          // objectFlags = none
mainFormTableID.w            // ID of above object
0.w                          // ID of below object
```

//Form Assignment **Using PilRC (version 2.9)**

HEX "fnav" ID mainFormID

// **Header**

0x00 0x01
0x00 0x02
0x00 0x14
0x00 0x08
0x00 0x00 0x00 0x01
0x03 0xE9
0x03 0xEA
0x03 0xEA
0x00 0x00

// **Objects in tab order**

0x03 0xE9
0x00 0x00
0x00 0x00
0x03 0xEA

0x03 0xEA
0x00 0x00
0x03 0xE9
0x00 0x00

//Form Assignment **USING Palm-RC (or PilRC 3.0 Beta +)**

HEX "fnav" ID mainFormID

// **Header**

1.w	// Resource version
2.w	// Objects in focus order
20.w	// Size of header
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// **Objects in tab order**

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Power Management

- **REM sleep**
  - New operating mode:
    - Device is running but the display is off!
  - REM sleep is initiated at the point when the device would otherwise go to deep sleep
    - (demo MP3 Player)

Power Management: REM Sleep

Normal System Sleep request and deferral

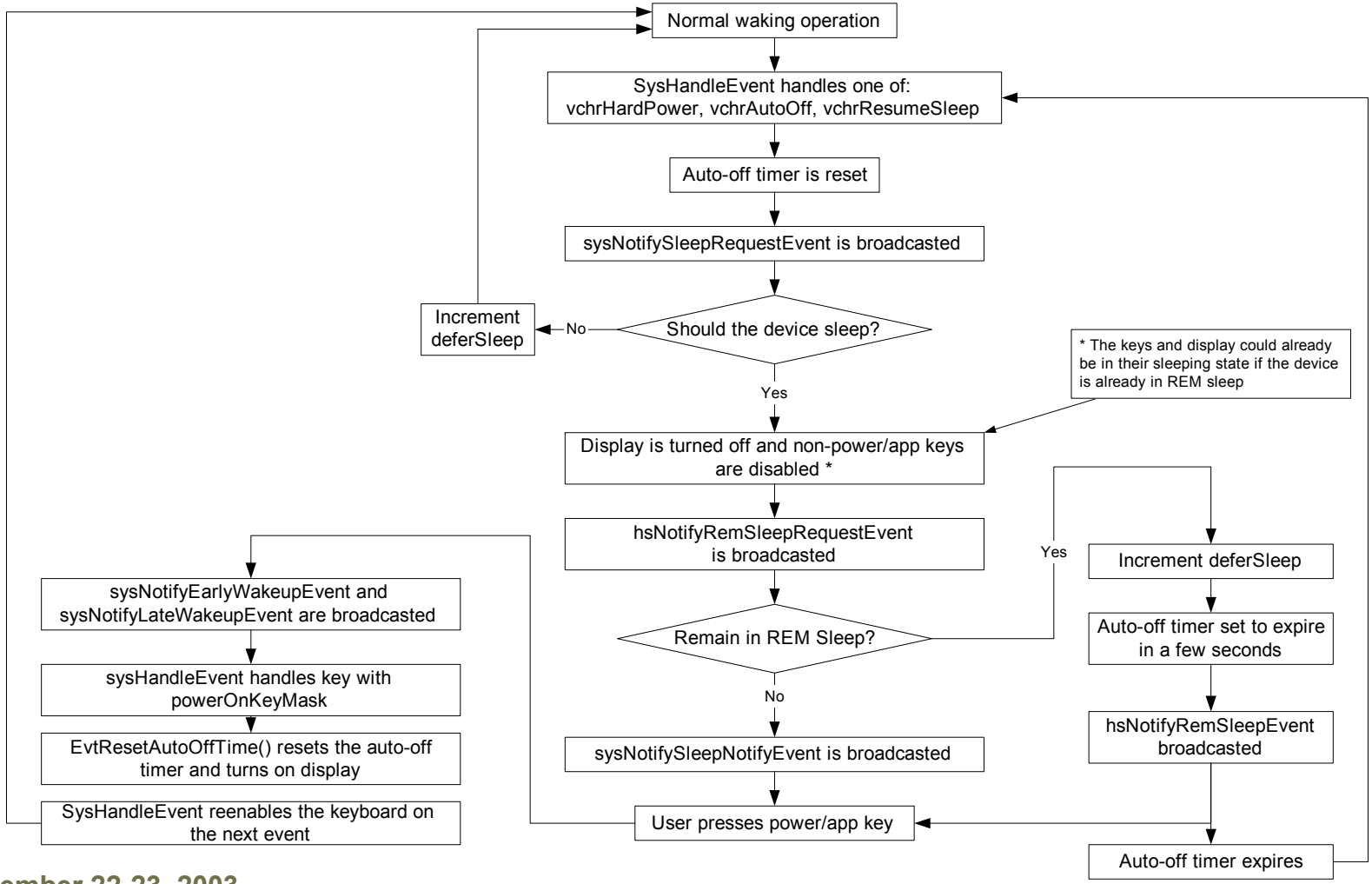
- **The device may be put to sleep for two reasons:**
  - The user pressed the power button (virtual key event = vchrHardPower)
  - The device is idle for the time set as the auto-off timeout. (virtual key event = vchrAutoOff)
  - Application gets sub-launch with sysNotifySleepRequestEvent
- **To stop the system sleep request**
  - Defer the Sleep request
    - Set the deferSleep member of the notification parameter block
- **If not stop**
  - Application would get notify with sysNotifySleepNotifyEvent

Power Management: REM Sleep

REM sleep usage

- **REM sleep happens between a sleep-request and a sleep notification**
  - Sleep-request notification is not deferred: display is turned off
  - Then a REM-request notification is broadcast.
- **If you need REM sleep:**
  - Sets deferSleep in the parameter block.
  - System broadcasts a REM notification indicating REM sleep has been entered.
- **A REM-request notification will be broadcasted until it is not deferred.**

Power Management: REM Sleep



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Telephony

- **Handspring Telephony Library**
  - Different than the standard Palm OS Telephony Library
  - Enables CDMA and GSM voice and data communication
- **Divided in 3 main categories**
  - GSM
  - CDMA
  - SMS
- **And then there's the events**
  - 40 of them!

- Data Connection
- GPRS or CDMA
- SIM Toolkit
- SMS, MMS & WAP

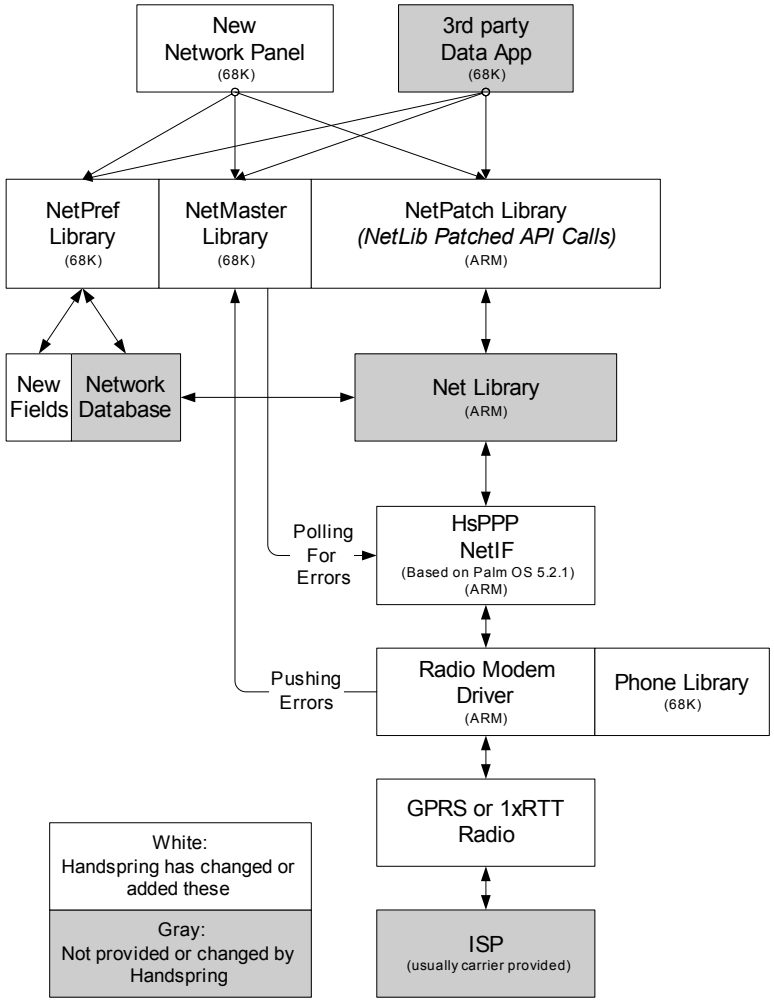
- Voice or Phone:
  - incoming calls,
  - disconnected calls,
  - …

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Network Connectivity

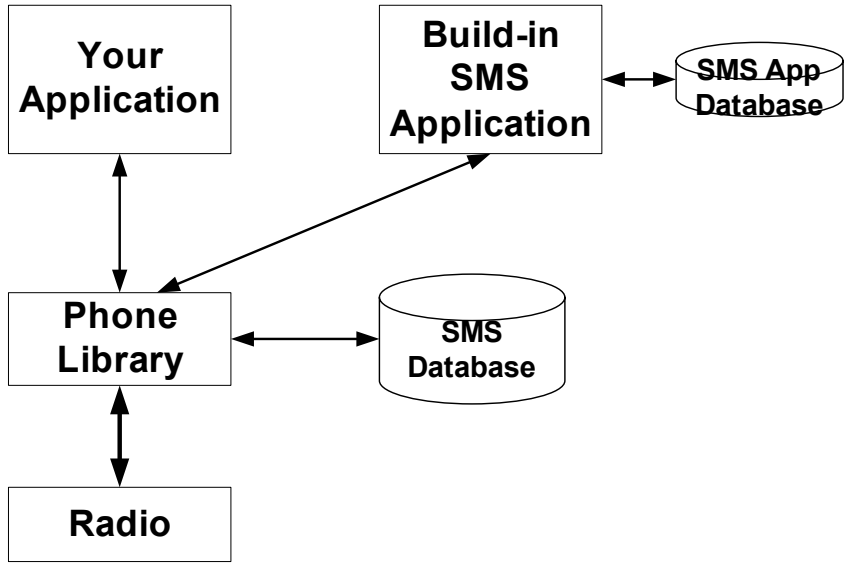
- **Patched Network Library**
  - Transfers control to NetMaster Library
- **NetMaster Library**
  - Packet data network management and control
  - Error messages from radio
  - Data Session Control API
    - Background login w/o any dialog box popup
    - Auto-dismiss dialog
    - Connection Arbitration (Priority, APN)
- **Network Preference Library**
  - Easily configure network services for each carrier



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  - **SMS & NBS Messages**

SMS & NBS



- **Use Phone Library to**
  - Receive and Decode Text Messages
  - Manage SMS messages
  - Do not go directly to SMS database!
- **How to use the library?**
  - Register with it:

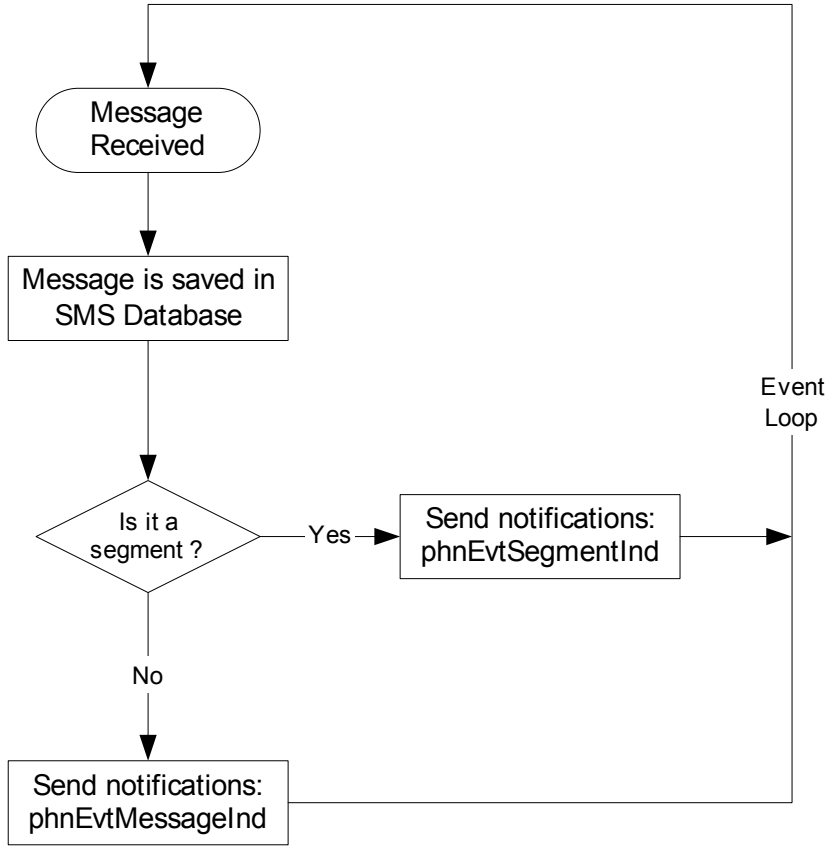
    HsGetPhoneLibrary (&PhoneLibRefNum);

    PhnLibRegister (libRef, appFileCreator, phnServiceSMS);
  - Un-register with the library when you're done!
- **You should use NBS messages**

  SysNotifyRegister (cardNo, dbID, phnNBSEvent, NULL, sysNotifyNormalPriority, NULL);
  - No conflict with build-in application
  - Header: //SCK<code>

SMS & NBS: incoming message

- **SMS:**
  - Message is saved in the Database by the Phone Library
  - Notification will be sent to registered application
  - Note: the Build-in SMS app pulls messages before anybody else if they are segments!
- **NBS:**
  - Message is part of the paramType when notification is received!



Q&A

- Summary:
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Thank You!